

Chapter 17

Curvature

DERIVATIVE OF ARC LENGTH. Let $y = f(x)$ be a function having a continuous first derivative. Let A (see Fig. 17-1) be a fixed point on the graph and denote by s the arc length measured from A to any other point on the curve. Let $P(x, y)$ be an arbitrary point and $Q(x + \Delta x, y + \Delta y)$ be a neighboring point on the curve. Denote by Δs the arc length from P to Q . The rate of change of the arc s ($= AP$) per unit change in x and the rate of change per unit change in y are given respectively by

$$\frac{ds}{dx} = \lim_{\Delta x \rightarrow 0} \frac{\Delta s}{\Delta x} = \pm \sqrt{1 + \left(\frac{dy}{dx}\right)^2}, \quad \frac{ds}{dy} = \lim_{\Delta y \rightarrow 0} \frac{\Delta s}{\Delta y} = \pm \sqrt{1 + \left(\frac{dx}{dy}\right)^2}$$

The plus or minus sign is to be taken in the first formula according as s increases or decreases as x increases, and in the second formula according as s increases or decreases as y increases.

When a curve is given by the parametric equations $x = f(u)$, $y = g(u)$, the rate of change of s with respect to u is given by $\frac{ds}{du} = \pm \sqrt{\left(\frac{dx}{du}\right)^2 + \left(\frac{dy}{du}\right)^2}$. Here the plus or minus sign is to be taken according as s increases or decreases as u increases.

To avoid the repetition of ambiguous signs, we shall assume hereafter that direction on each arc considered has been established so that the derivative of arc length will be positive.

See Problems 1-5.

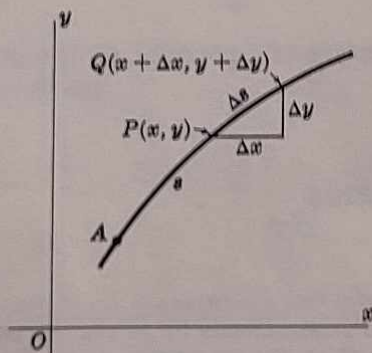


Fig. 17-1

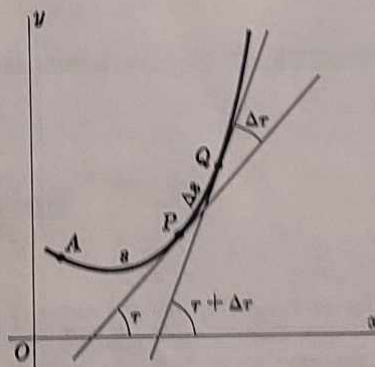
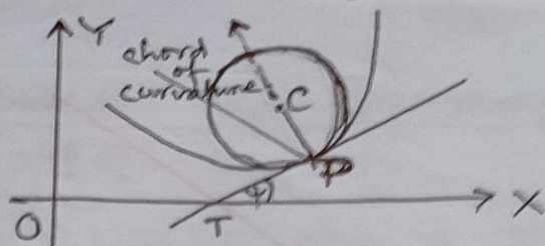


Fig. 17-2

CURVATURE. The curvature K of a curve $y = f(x)$, at any point P on it, is the rate of change in direction (i.e., the angle of inclination τ of the tangent line at P) per unit of arc length s . (See Fig. 17-2.) Thus,

$$K = \frac{d\tau}{ds} = \lim_{\Delta s \rightarrow 0} \frac{\Delta\tau}{\Delta s} = \frac{\frac{d^2y}{dx^2}}{\left\{1 + \left(\frac{dy}{dx}\right)^2\right\}^{3/2}}; \quad K = \frac{-\frac{d^2x}{dy^2}}{\left\{1 + \left(\frac{dx}{dy}\right)^2\right\}^{3/2}}$$



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If a length $P = PC$ is measured from a fixed point P along the positive direction of the normal, the point C is called the centre of curvature at P, and the circle with centre at C and radius CP is called the circle of curvature at P.

Any chord of this circle through the point of contact is called a chord of curvature.

Formulae for radius of curvature:

(A) For the Cartesian equation $y = f(x)$.

We know $\frac{dy}{dx} = \tan \psi \Rightarrow \frac{dy/ds}{dx/ds} = \frac{\sin \psi}{\cos \psi}$.

$$\therefore \frac{d^2y}{dx^2} = \sec^2 \psi \frac{d\psi}{dx} = \frac{d}{dx} \left(\frac{dy}{dx} \right) = \frac{d}{d\psi} (\tan \psi) \frac{d\psi}{dx}$$

$$= \sec^2 \psi \frac{d\psi}{ds} \cdot \frac{ds}{dx} = \sec^3 \psi \frac{d\psi}{ds}$$

$$\therefore P = \frac{ds}{d\psi} = \frac{\sec^3 \psi}{\frac{d^2y}{dx^2}} \text{ We have } \sec \psi = (1 + \tan^2 \psi)^{1/2}$$

$$= \left\{ 1 + \left(\frac{dy}{dx} \right)^2 \right\}^{1/2}$$

$$\text{Thus, } P = \frac{\left\{ 1 + \left(\frac{dy}{dx} \right)^2 \right\}^{3/2}}{\frac{d^2y}{dx^2}} = \frac{(1 + y_1^2)^{3/2}}{y_2} \quad \dots (1) \text{ if } y_2 \neq 0$$

Similarly, if $x = \phi(y)$, the radius of curvature is

$$P = \frac{(1 + x_1^2)^{3/2}}{x_2} \text{ if } x_2 \neq 0 \quad \dots (2)$$

Example 1. Find the radius of curvature of the parabola $y^2 = 4x$ at $(0,0)$.

Solution: Here, the y -axis is the tangent to the parabola $y^2 = 4x$ at $(0,0)$.

We have, $x = \frac{1}{4}y^2 = \phi(y)$.

$$\therefore \frac{dx}{dy} = \frac{1}{2}y \quad \text{and} \quad \frac{d^2x}{dy^2} = \frac{1}{2}.$$

At $(0,0)$, $\kappa_1 = \frac{dx}{dy} = 0$ and $\kappa_2 = \frac{d^2x}{dy^2} = \frac{1}{2}$.

$\therefore$ at $(0,0)$ the radius of curvature,

$$\begin{aligned} \rho &= \frac{(1+\kappa_1^2)^{3/2}}{\kappa_2} \\ &= \frac{(1+0)^{3/2}}{1/2} = \left(\frac{1}{2}\right)^{-1} = 2. \end{aligned}$$

Example 2. Find the radius of curvature at the origin of the conic $y - x = x^2 + 2xy + y^2$.

Solution: We can write

$$y_1 - 1 = 2x + 2xy_1 + 2y + 2yy_1$$

$$y_2 = 2 + 2xy_2 + 2y_1 + 2y_1 + 2yy_2 + 2y_1^2$$

Now, at $(0,0)$, $y_1 = 1$ and $y_2 = 8$.

$$\begin{aligned} \therefore \text{at the origin, } \rho &= \frac{(1+y_1^2)^{3/2}}{y_2} = \frac{(1+1)^{3/2}}{8} = \frac{2\sqrt{2}}{8} \\ &= \frac{\sqrt{2}}{4} = .35 \text{ (approximately)} \end{aligned}$$

Example 3. For any curve, prove that $\frac{1}{\rho^2} = \left(\frac{d^2x}{ds^2}\right)^2 + \left(\frac{d^2y}{ds^2}\right)^2$.

Solution: We have, $\frac{dx}{ds} = \cos \psi$

$$\therefore \frac{d^2x}{ds^2} = -\sin \psi \cdot \frac{d\psi}{ds} = -\sin \psi \cdot \frac{1}{\rho}$$

(1)

Also, $\frac{dy}{ds} = \sin \psi$ and $\frac{d^2y}{ds^2} = \cos \psi \frac{d\psi}{ds}$
 $= \cos \psi \cdot \frac{1}{\rho} \dots (2)$

Thus, from (1) and (2),

$$\frac{1}{\rho^2} = \left(\frac{d^2x}{ds^2} \right)^2 + \left(\frac{d^2y}{ds^2} \right)^2$$

Centre of curvature

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Let $(\bar{x}, \bar{y})$ be the coordinates of the centre of curvature C corresponding to any point $P(x, y)$ on the curve.

Since $C(\bar{x}, \bar{y})$ lies on the normal ~~line~~, $(\bar{x} - x) + (\bar{y} - y)y_1 = 0$, we get

$$(\bar{x} - x) + (\bar{y} - y)y_1 = 0 \quad \dots (1)$$

Again, since $PC = \rho$, i.e., $\rho^2 = \frac{(1+y_1^2)^3}{y_2^2}$,
we have $(\bar{x} - x)^2 + (\bar{y} - y)^2 = \rho^2 = \frac{(1+y_1^2)^3}{y_2^2}$
 $\dots (2)$

From (1), we can write

$$\bar{x} - x = -(\bar{y} - y)y_1 \quad \dots (3)$$

$\therefore$ (2) becomes,

$$(\bar{y} - y)^2 y_1^2 + (\bar{y} - y)^2 = \frac{(1+y_1^2)^3}{y_2^2}$$

$$\Rightarrow (\bar{y} - y)^2 (1+y_1^2) = \frac{(1+y_1^2)^3}{y_2^2} \quad \dots (4)$$

$$\text{or } (\bar{y} - y)^2 = \frac{(1+y_1^2)^2}{y_2^2}$$

$$\Rightarrow \bar{y} - y = \frac{1+y_1^2}{y_2} \quad \dots (5)$$

$$\text{Again, } \bar{x} - x = -(\bar{y} - y)y_1 = -\frac{1+y_1^2}{y_2} \times y_1$$

$$\therefore \bar{x} - x = -\frac{y_1(1+y_1^2)}{y_2} \quad \dots (6)$$

Thus, from (5) and (6), we get

$$\bar{x} = x - \frac{y_1(1+y_1^2)}{y_2}, \quad \bar{y} = y + \frac{1+y_1^2}{y_2}$$

Cor. Hence the eqn. of the circle of curvature is
 $(x - \bar{x})^2 + (y - \bar{y})^2 = \rho^2$

Example 1. Find the centre of curvature at any point (x, y) on the parabola $y^2 = 4ax$.

Solution: Here $2yy_1 = 4a \Rightarrow y_1 = \frac{2a}{y} = \frac{2a}{2\sqrt{ax}}$
 $\Rightarrow y_1 = \sqrt{\frac{a}{x}}$ and $y_2 = \frac{1}{2} \left(\frac{a}{x} \right)^{-1/2} = -\frac{a}{x^{3/2}}$
 $= -\frac{1}{2} \cdot \frac{\sqrt{a}}{x^{3/2}}$

$$\therefore y_1(1+y_1^2) = \sqrt{\frac{a}{x}} \left(1 + \frac{a}{x} \right) = \frac{\sqrt{a}}{x^{3/2}} (x+a).$$

If $(\bar{x}, \bar{y})$ be the centre of curvature, we have $\bar{x} = x - \frac{y_1(1+y_1^2)}{y_2} = x + \frac{\frac{\sqrt{a}}{x^{3/2}}(x+a)}{-\frac{1}{2} \cdot \frac{\sqrt{a}}{x^{3/2}}}$

$$= x + 2(x+a) = 3x+2a$$

and $\bar{y} = y + \frac{1+y_1^2}{y_2} = y + \frac{1 + \frac{a}{x}}{-\frac{1}{2} \cdot \frac{\sqrt{a}}{x^{3/2}}}$

$$= y - \frac{2\sqrt{x}(x+a)}{\sqrt{a}}$$

$$= 2\sqrt{ax} - \frac{2\sqrt{x}(x+a)}{\sqrt{a}} = \frac{2a\sqrt{x} - 2x\sqrt{x} - 2a\sqrt{x}}{\sqrt{a}}$$

$$= -\frac{2x^{3/2}}{\sqrt{a}}$$

Example 2. Find the equation of the circle of curvature at the point $(3, 1)$ on the curve $y = x^2 - 6x + 10$.

Solution: Here, $y_1 = 2x - 6$ and $y_2 = 2$.

$\therefore$ at $(3, 1)$, $y_1 = 0$ and $y_2 = 2$.

We have $\rho = \frac{(1+y_1^2)^{3/2}}{y_2} = \frac{(1+0)^{3/2}}{2} = \frac{1}{2}$, at $(3, 1)$

Let $(\bar{x}, \bar{y})$ be the centre of curvature. Then at $(3, 1)$, we have

$$\bar{x} = x - \frac{y_1(1+y_1^2)}{y_2} = 3 - 0 = 3,$$

$$\bar{y} = y + \frac{1+y_1^2}{y_2} = 1 + \frac{1}{2} = \frac{3}{2}.$$

$\therefore$ The equation of the required circle of curvature is

$$(x-3)^2 + (y-\frac{3}{2})^2 = (\frac{1}{2})^2$$

$$\Rightarrow x^2 - 6x + 9 + y^2 - 3y + \frac{9}{4} = \frac{1}{4}$$

$$\Rightarrow x^2 + y^2 - 6x - 3y + 11 = 0.$$

Example 3. Find the equation of the circle of curvature of $x^3 + y^3 = 3axy$ at $(\frac{3a}{2}, \frac{3a}{2})$.

Solution: We have $f(x, y) = x^3 + y^3 - 3axy = 0$.

$$\Rightarrow 3x^2 + 3y^2 y_1 - 3axy_1 - 3ay = 0$$

$$\Rightarrow x^2 + y^2 y_1 - axy_1 - ay = 0 \quad \dots (1)$$

At $(\frac{3a}{2}, \frac{3a}{2})$,

$$(1) \Rightarrow \frac{9a^2}{4} + \frac{9a^2}{4} y_1 - \frac{3a^2}{2} y_1 - \frac{3a^2}{2} = 0$$

$$\Rightarrow \left(\frac{9a^2}{4} - \frac{3a^2}{2}\right) y_1 + \left(\frac{9a^2}{4} - \frac{3a^2}{2}\right) = 0$$

$$\Rightarrow y_1 = -1.$$

Again, $(1) \Rightarrow 2x + 2y y_1^2 + y^2 y_2 - axy_2 - 2ay_1 = 0$

$$\Rightarrow 2 \cdot \frac{3a}{2} + 2 \cdot \frac{3a}{2} (-1)^2 + \frac{9a^2}{4} y_2 - \frac{3a^2}{2} y_2 + 2a = 0$$

at $(\frac{3a}{2}, \frac{3a}{2})$

$$\Rightarrow 8a + \frac{3a^2}{4} y_2 = 0$$

$$\therefore y_2 = -\frac{32}{3a}.$$

We have, radius of curvature, $\rho = \frac{(1+y_1^2)^{\frac{3}{2}}}{y_2}$

$$= \frac{2\sqrt{2}}{-\frac{32}{3a}} = -\frac{3a}{8\sqrt{2}}.$$

Let $(\bar{x}, \bar{y})$ be the centre of curvature. 8
Then, at $(\frac{3a}{2}, \frac{3a}{2})$,

$$\bar{x} = x - \frac{y_1(1+y_1^2)}{y_2^2} = \frac{3a}{2} - \frac{(-1)(1+1)}{-\frac{32}{3a}}$$

$$= \frac{3a}{2} - \frac{3a}{16} = \frac{21a}{16}$$

$$\bar{y} = y + \frac{1+y_1^2}{y_2} = \frac{3a}{2} + \frac{2}{-\frac{32}{3a}} = \frac{21a}{16}$$

Thus, the equation of the circle of curvature at $(\frac{3a}{2}, \frac{3a}{2})$ is

$$\left(x - \frac{21a}{16}\right)^2 + \left(y - \frac{21a}{16}\right)^2 = \left(-\frac{3a}{8\sqrt{2}}\right)^2$$

$$\Rightarrow x^2 + y^2 - \frac{21}{8}ax - \frac{21}{8}ay + \frac{441}{128}a^2 = \frac{9a^2}{128}$$

$$\Rightarrow 8x^2 + 8y^2 - 21ax - 21ay - 27a^2 = 0.$$